

paragraph (but extending it to the full cutoff space $(b^*1, b^{**}1)$), we conclude that the cutoffs in the most-sharing equilibrium of the complete sharing network must satisfy $(b^*1, b^{**}1) \preceq (b^{*'}, b^{**'})$, and in particular, $b^*1 \preceq b^{*'} = 0$. This implies that all agents share or ignore in the complete network, yielding a contradiction. Therefore, such an interval $(0, \bar{b})$ must exist, and in particular, we choose the largest such $\bar{b}$ (in the supremum sense) where agents with priors in $(0, \bar{b})$ *never* share in any sharing network $\mathbf{P}$ in the most-sharing equilibrium.

Next, we consider disconnecting (and removing) all agents in any community ℓ with $b^{(\ell)} < \bar{b}$, but leaving all other communities connected in a (partial) complete network. We call this network the *active network*. We claim that when ε is sufficiently small, all of the remaining agents in the active network either share or ignore. By definition, an agent with $\bar{b}$ would share under some sharing network $\mathbf{P}^*$ but any agent with $\bar{b} - \epsilon$ would ignore (for arbitrarily small ϵ) under $\mathbf{P}^*$ (and by leveraging Lemma A.4(i), not all agents with $b < \bar{b}$ dislike). This implies that $U_i(\mathcal{D}) < 0$ for an agent with prior $\bar{b}$ (by monotonicity), and thus an agent with this prior either shares or ignores in the active network. Moreover, for agents with priors in a small half-open neighborhood around $\bar{b}$ (i.e., an interval $(\bar{b} - \eta, \bar{b}]$ for some $\eta > 0$) ignoring is a better response to disliking in the active network. Thus, for sufficiently small ε , we obtain a partial complete network (the active network) with agents who only share and ignore (with probability 1) and never dislike (with probability 0).

Finally, with these two observations, we claim that the profit-maximizing sharing network takes the form of part (i). First, consider all communities who participated in the active network described above (call these the active communities) and suppose that the agents in communities outside of this active network are non-existent in our model (call these the inactive communities). When the active communities are arranged in a complete sharing network, we showed in the previous paragraph that all agents either share or ignore. By the exact argument in the first two paragraphs then, engagement (and virality) are maximized (amongst only the active communities) when these communities are arranged in a complete sharing network. Second, by construction of the active network (and the active communities), all agents in inactive communities *never* share under *any* sharing network $\mathbf{P}$. Therefore, removing these agents is without loss to potential virality. Hence, whenever virality is maximized amongst only agents in active communities, it is also maximized in general.

Lastly, we note that we can form a (partial) complete network among the inactive communities, but provide no connections to the (partial) complete network of active communities who only share and ignore (with probability 1). By our previous observations, this is a profit-maximizing sharing network for the platform. At the same time, it is exactly the form of a two-island model with $(p_s, p_d) = (1, 0)$, which has maximal homophily. ■

Proof of Proposition 4. Note by Theorem 3 that an agent i with prior $b_i = b^{(k+1)}$ is indifferent between ignoring and disliking when $r = r_P$ (but strictly prefers to either share or ignore for all $r > r_P$), so r_P increases if and only if this agent (strictly) prefers to dislike following a shift in parameters. Because $b^{(k+1)} < 1/2$, an increase in polarization means that agent i 's prior decreases (see Lemma A.3), and